

$$1. |\vec{v}_1\rangle = 2|\uparrow\rangle + 1|\downarrow\rangle$$

$$= \frac{2|\uparrow\rangle + 1|\downarrow\rangle}{\sqrt{2^2 + 1^2}}$$

$$= \frac{2}{\sqrt{5}}|\uparrow\rangle + \frac{1}{\sqrt{5}}|\downarrow\rangle$$

$$3. |\vec{v}_1\rangle = 2|\uparrow\rangle + (1+i)|\downarrow\rangle$$

$$= \frac{2}{\sqrt{2^2 + (1-i)(1+i)}}|\uparrow\rangle + \frac{(1+i)}{\sqrt{2^2 + (1-i)(1+i)}}|\downarrow\rangle$$

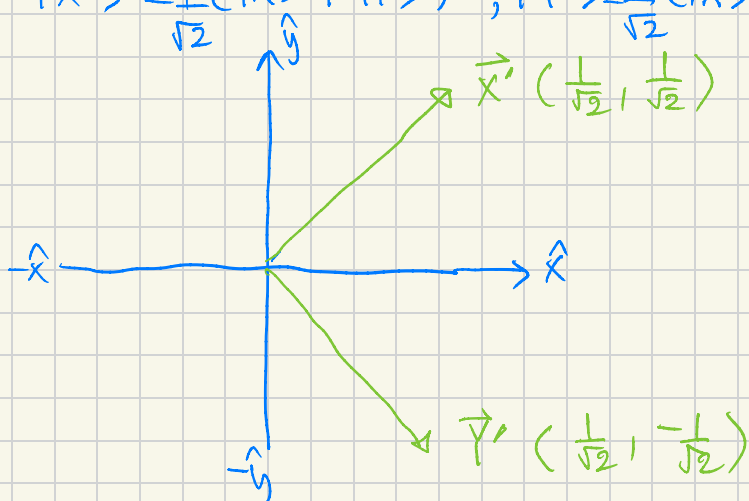
$$= \frac{2}{\sqrt{4 + (1^2 - i^2)}}|\uparrow\rangle + \frac{(1+i)}{\sqrt{4 + (1^2 - i^2)}}|\downarrow\rangle$$

$$= \frac{2}{\sqrt{4+0}}|\uparrow\rangle + \frac{(1+i)}{\sqrt{4+0}}|\downarrow\rangle$$

$$= \frac{2}{\sqrt{6}}|\uparrow\rangle + \frac{(1+i)}{\sqrt{6}}|\downarrow\rangle$$

$$= (0.816 + 0j, 0.4082 + 0.408j)$$

$$5. |\vec{x}'\rangle = \frac{1}{\sqrt{2}}(|\hat{x}\rangle + |\hat{y}\rangle), |\vec{y}'\rangle = \frac{1}{\sqrt{2}}(|\hat{x}\rangle - |\hat{y}\rangle)$$



$$\langle \vec{x}' | \vec{x}' \rangle = \left(\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \right) \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{2} + \frac{1}{2} = 1 \text{ (Normalized)}$$

$$\langle \vec{y}' | \vec{y}' \rangle = \left(\frac{1}{\sqrt{2}} \quad -\frac{1}{\sqrt{2}} \right) \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{2} + \frac{1}{2} = 1 \longrightarrow$$

$$\langle \vec{x}' | \vec{y}' \rangle = \left(\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \right) \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{2} - \frac{1}{2} = 0 \text{ Orthogonal}$$

$$\langle \vec{y}' | \vec{x}' \rangle = \left(\frac{1}{\sqrt{2}} \quad -\frac{1}{\sqrt{2}} \right) \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{2} - \frac{1}{2} = 0 \longrightarrow$$

6. $\vec{V}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}_{|+\rangle/|-\rangle}$

$$|0\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle \quad \text{or} \quad |1\rangle = \frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle \quad \text{--- ①}$$

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{or} \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \quad \text{--- ②}$$

$$\vec{V}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \text{ in } |+\rangle/|-\rangle \text{ basis}$$

$$\text{We get } |0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + 2|-\rangle) \quad \text{or} \quad |1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - 2|-\rangle) \quad \text{--- ③}$$

Converting back by substituting ③ in ②

$$\begin{aligned} |+\rangle &= \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}(2|+\rangle) \right) = 1 \\ |-\rangle &= \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}(4|-\rangle) \right) = 2 \end{aligned} \quad \left. \vphantom{\begin{aligned} |+\rangle \\ |-\rangle \end{aligned}} \right\} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

7. $\vec{V}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}_{|0\rangle/|1\rangle}$

$$|\vec{V}_1\rangle = |10\rangle + 2|11\rangle$$

converting it to $|+\rangle, |-\rangle$ basis

$$\begin{aligned} &= \left(\frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle \right) + 2 \left(\frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle \right) \\ &= \frac{3}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle \end{aligned}$$

Normalized values are:

$$\begin{aligned} &\left(\frac{\frac{3}{\sqrt{2}}}{\sqrt{\left(\frac{3}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2}}, \frac{-\frac{1}{\sqrt{2}}}{\sqrt{\left(\frac{3}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2}} \right) = \left(\frac{\frac{3}{\sqrt{2}}}{\sqrt{\frac{10}{2}}}, \frac{-\frac{1}{\sqrt{2}}}{\sqrt{\frac{10}{2}}} \right) = \frac{3}{\sqrt{10}}, \frac{-1}{\sqrt{10}} \\ &= (0.948, -0.316) \end{aligned}$$

8. $|\vec{V}_1\rangle = |10\rangle + 2|11\rangle$

$$\begin{aligned} &= \left(\frac{1}{\sqrt{1+4}}, \frac{2}{\sqrt{1+4}} \right) = \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right) \\ &= (0.447, 0.894) \end{aligned}$$